

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM1-5-FINAL | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B083-Networkwalks
Date	15 September 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM2 (GHDB) W2-PM3 (Maltego) W2-PM4 (theHarvester) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. Microsoft.com (accessing publicly available information) 3. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This final Week 2 report consolidates the five completed project modules: multiple Kali Linux footprinting tools (W2-PM1), Google Hacking Database reconnaissance (W2-PM2), Maltego footprinting (W2-PM3), theHarvester reconnaissance against microsoft.com (W2-PM4), and Zenmap network discovery on my local LAN (W2-PM5).

The activities demonstrate a progression from passive information gathering to network discovery. Work against networkwalks.com was performed with the required permission, Microsoft.com information was gathered from public sources as specified by the practical, and Zenmap scanning was performed against my own local network. Screenshots have been added in the evidence section of this report.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification
GHDB (Google Hacking Database)	Use Google dorks to identify publicly indexed exposed devices and files.
Maltego	Map relationships around a domain and run email-related transforms.
theHarvester	Gather publicly available email addresses, sub-domains and hosts from multiple sources.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I completed W2-PM1 by performing passive reconnaissance of networkwalks.com with six Kali Linux tools. WHOIS returned domain registration and name-server information; WhatWeb identified the web stack; Nslookup resolved the domain; Curl inspected HTTP response headers; Wafw00f identified the WAF; and DNSRecon enumerated DNS infrastructure.

WHOIS identified GoDaddy as the registrar, a 2019 creation date, a 2027 expiry date, and HostGator name servers. The registrant information was privacy-protected. WhatWeb identified Apache, Bootstrap

7.1, jQuery 3.7.1, WordPress 7.1 and WordPress Download Manager 3.3.58, with the site resolving to 192.232.216.135.

Curl returned HTTP/2 200 response headers and exposed the WordPress REST API reference /wp-json/. Wafw00f identified ModSecurity (SpiderLabs), while DNSRecon returned SOA, NS, MX, A, TXT and SRV records and reported unsigned DNSSEC.

4.2 Footprinting & Reconnaissance with GHDB

W2-PM2 used the Google Hacking Database (GHDB) and Google search operators to identify publicly indexed exposures. The practical required two searches: ten live exposed security-camera interfaces and ten listings containing downloadable mathematics PDF material.

For Task 1, the completed worksheet recorded ten exposed camera endpoints and their associated Google dorks, with no usernames or passwords recorded. Because these were live third-party camera interfaces, the final report intentionally records the finding and methodology without reproducing the live access links.

For Task 2, ten publicly indexed mathematics PDF directory listings were recorded. The exercise demonstrated how search-engine indexing can expose directory structures and documents without directly interacting with a target system.

4.3 Footprinting with Maltego

W2-PM3 involved downloading and installing Maltego on Windows, completing the initial configuration and creating the free Maltego account. A Domain entity was then created for networkwalks.com and email-related transforms were run to identify email addresses associated with the organization.

Evidence below confirmed that email-related entities were harvested from the domain. The result provides useful reconnaissance context because exposed organizational email addresses can contribute to later social-engineering and phishing risk assessments.

4.4 Footprinting & Reconnaissance with theHarvester

For W2-PM4, I used theHarvester 4.8.2 against microsoft.com. Task 1 used the command "theHarvester -d microsoft.com -l 1000 -b baidu". The run returned 0 emails address and 7 hosts, with no IPs or people reported.

Task 2 used "theHarvester -d microsoft.com -l 50 -b all". The run returned 7 email addresses, no IPs and no people, and reported 1,312 hosts. Several data sources could not be used because API keys were missing or source connections failed; the run also reported an API-endpoint wordlist error. These limitations were documented rather than treated as findings.

The results demonstrate the scale of information that can be assembled from public sources. TheHarvester documentation also notes that results can change as external information sources change, so the output represents the state observed during this task.

4.5 Network Scanning with Zenmap

W2-PM5, I used Zenmap, the Nmap graphical interface, to perform a Ping Scan of the local 10.0.0.0/24 subnet. The completed scan reported four live hosts.

The observed live hosts were 10.0.0.1, 10.0.0.2, 10.0.0.9 and 10.0.0.10. The scan output recorded the MAC addresses 52:54:00:12:35:00 (QEMU virtual NIC), 08:00:27:A7:5B:C5 (Oracle VirtualBox virtual

NIC), 08:00:27:FF:36:16 (Oracle VirtualBox virtual NIC), and 08-00-27-E7-F1-2D for the local machine. The scan covered 256 IP addresses and completed with four hosts up.

The Zenmap Topology view was also used as required by the task, with the legend enabled and the topology saved as a PDF.

5. Risk Analysis / Impact

Based on the combined results from W2-PM1 through W2-PM5, the following are potential security risks and observations. They represent information exposure and discovery findings, not confirmed vulnerabilities.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version information to identify software requiring further security review	● Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● Low
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● Medium
6	Multiple live hosts visible on local network	Zenmap identified four live hosts on 10.0.0.0/24	Unknown devices may increase the network attack surface	● Medium
7	Publicly indexed camera interfaces	GHDB identified 10 exposed camera endpoints	Potential privacy, access-control and security risk	● High
8	Public directory/file exposure	GHDB identified 10 mathematics PDF directory listings	May reveal directory structure or files	● Low
9	Organizational email addresses exposed	Maltego returned email entities for networkwalks.com	May support phishing and social engineering	● Medium
10	Large public sub-domain/host footprint	theHarvester returned 1,312 hosts	Large public host footprint requires asset review	● Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**
Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
2. **Keep software updated**
CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers**
HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly**
DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF**
Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery**
Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**
Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**
Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.
10. **Review Google-indexed exposure**
Organizations should periodically search their own domains for exposed devices, directories and documents and remove or restrict information that should not be public.
11. **Secure externally accessible surveillance devices**
Camera interfaces should require strong authentication, appropriate network restrictions and secure configuration rather than being directly exposed to the public internet.
12. **Minimize public email exposure**
Public-facing contact addresses should be separated from sensitive operational accounts and protected with appropriate anti-phishing controls.

13. **Maintain an accurate sub-domain inventory**

Large externally visible host lists should be reviewed so that unused, forgotten or unnecessary sub-domains can be removed or restricted.

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Large externally visible host lists should be reviewed so that unused, forgotten or unnecessary sub-domains can be removed or restricted.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed five practical modules covering passive footprinting, Google-based reconnaissance, relationship mapping, public-source harvesting and local network discovery.

Across the modules, I used WHOIS, WhatWeb, Nslookup, Curl, Wafw00f, DNSRecon, GHDB, Maltego, theHarvester and Zenmap. These tools demonstrated how publicly available information can reveal domain registration details, technologies, DNS infrastructure, indexed exposures, organizational email information, public hosts and active local devices.

TheHarvester showed how results can vary by source and how missing API keys or source failures can affect collection. Zenmap then provided a practical view of local network discovery by identifying four live hosts and their MAC addresses on the scanned subnet.

The combined exercises showed that reconnaissance is a major part of cybersecurity assessment. Before exploitation is considered, a security professional can build an understanding of the visible attack surface and identify information that defenders may need to reduce or monitor.

I also learned that technical findings must be documented with their evidence, limitations and potential impact. Public exposure does not automatically mean a confirmed vulnerability; further authorized validation is required before assigning a confirmed vulnerability status.

Finally, all reconnaissance and scanning activities must remain within an authorized scope. The Networkwalks practicals were completed for educational purposes, with Networkwalks used under the stated permission, Microsoft.com information gathered from public sources as specified by the task, and Zenmap used on my own local network.

8. Evidences Collected

Figure 1 – W2-PM1: WHOIS and WhatWeb output.

```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-15T08:25:09Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for

kali@kali: ~
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[[_wpdm_client], Country[UNITED STATES][0], HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Country[UNITED STATES][0], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPID&pidnVar2=58511&pidnVar2=76&pidnVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1], Country[UNITED STATES][0], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPID&pidnVar2=58511&pidnVar2=76&pidnVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
```

Figure 2 – W2-PM1: Nslookup, Curl and Wafw00f output.

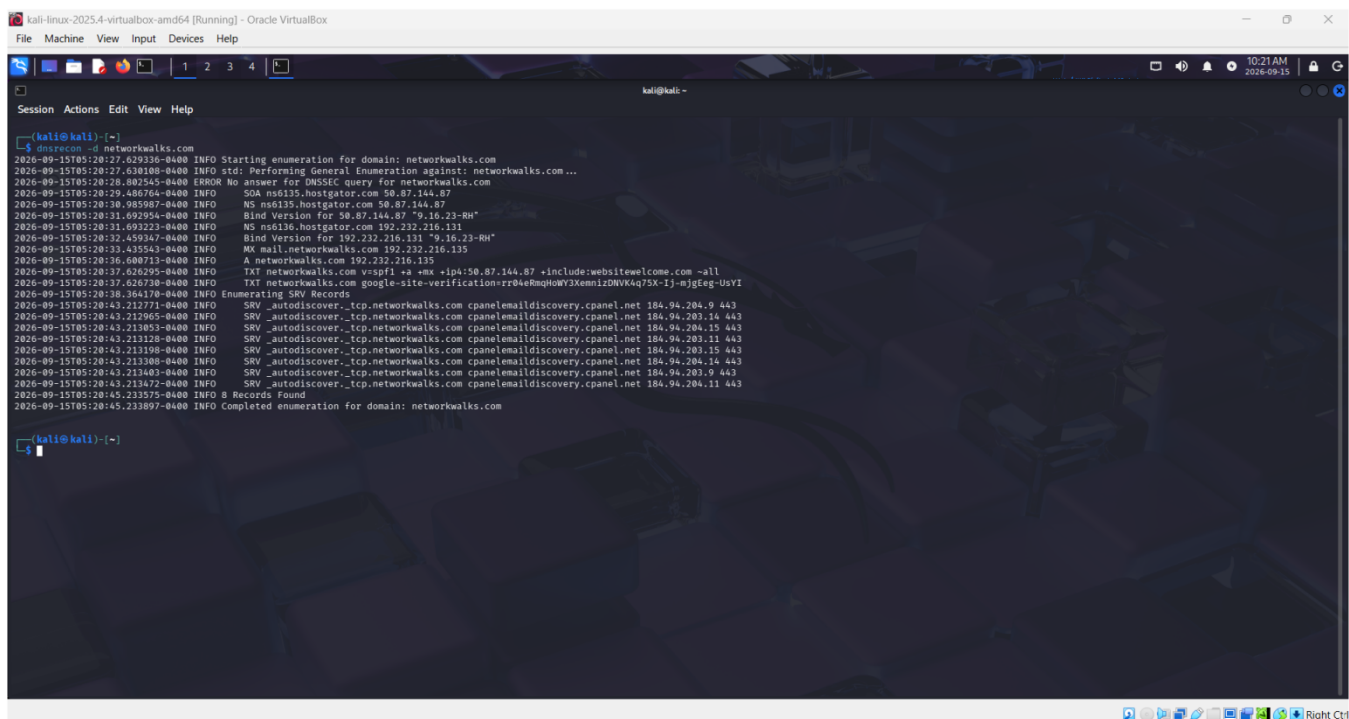


Figure 4 – W2-PM2: GHDB search and completed findings evidence.

Date Added	Dork	Category	Author
2023-11-07	intitle:"Webcam" inurl:WebCam.htm	Various Online Devices	s Thakur
2023-03-14	Google Dork : intitle:"Toshiba Network Camera"	Various Online Devices	Shilpa Shaji
2023-02-27	intitle:"Index of /cam/"	Files Containing Juicy Info	Sanu Jose M
2023-01-31	intitle:"Index of /webcam/"	Files Containing Juicy Info	Shuvrosayar Das
2022-07-25	intitle:"Device(IP CAMERA)" language:-com:net	Pages Containing Login Portals	s Thakur
2022-07-25	intitle:"NoVus IP camera" -com	Various Online Devices	s Thakur
2022-07-20	intitle:"Network Camera" inurl:main.cgi	Various Online Devices	s Thakur
2022-06-17	inurl:webcam site:skylinewebcams.com inurl:roma	Various Online Devices	Simone Gasparato
2021-11-16	intitle:"Login" intext:"cam"	Pages Containing Login Portals	Yash Singh
2021-11-16	intext:"Real-time IP Camera Monitoring System" intext:"ActiveX Mode (For IE Browser)"	Various Online Devices	Yash Singh
2021-11-12	intitle:"Login" intext:"camera"	Pages Containing Login Portals	Naveen Venugopal
2021-11-09	intitle:"webcamXP" inurl:8080	Various Online Devices	Krishna Agarwal
2021-11-09	intitle:"Index of" "DCIM/camera"	Files Containing Juicy Info	Krishna Agarwal
2021-10-28	intitle:ip camera login page	Pages Containing Login Portals	jawhar milkan
2021-10-26	intitle:"Microseven M7CAM IP Camera"	Various Online Devices	Neha Singh

Tasks

Task1 Find 10x live vulnerable security camera links that are exposed & accessible from Internet. List them in below table along with their relevant dork that you have used.

No.	Link	Relevant Dork	Username /Password (if any)
1	http://122.116.41.8:8080/	intitle:"webcamXP" inurl:8080	---
2	http://109.233.191.130:8080/	intitle:"webcamXP 5" -download	---
3	http://83.41.12.44/	intitle:"webcamXP 5" -download	---
4	http://75.149.26.30:1024/	intitle:"webcamXP 5" -download	---
5	http://95.255.183.164:8080/	intitle:"webcamXP" inurl:8080	---
6	http://85.93.53.175:8080/	intitle:"webcamXP" inurl:8080	---
7	http://109.233.191.226:8090/	intitle:"webcamXP" inurl:8080	---
8	http://99.114.240.169:8080/	intitle:"webcamXP" inurl:8080	---
9	http://109.206.96.75:8080/	intitle:"webcamXP" inurl:8080	---
10	http://68.115.218.130:32479/	intitle:"webcamXP" inurl:8080	---

Figure 5 – W2-PM3: Maltego domain entity and email transform results.

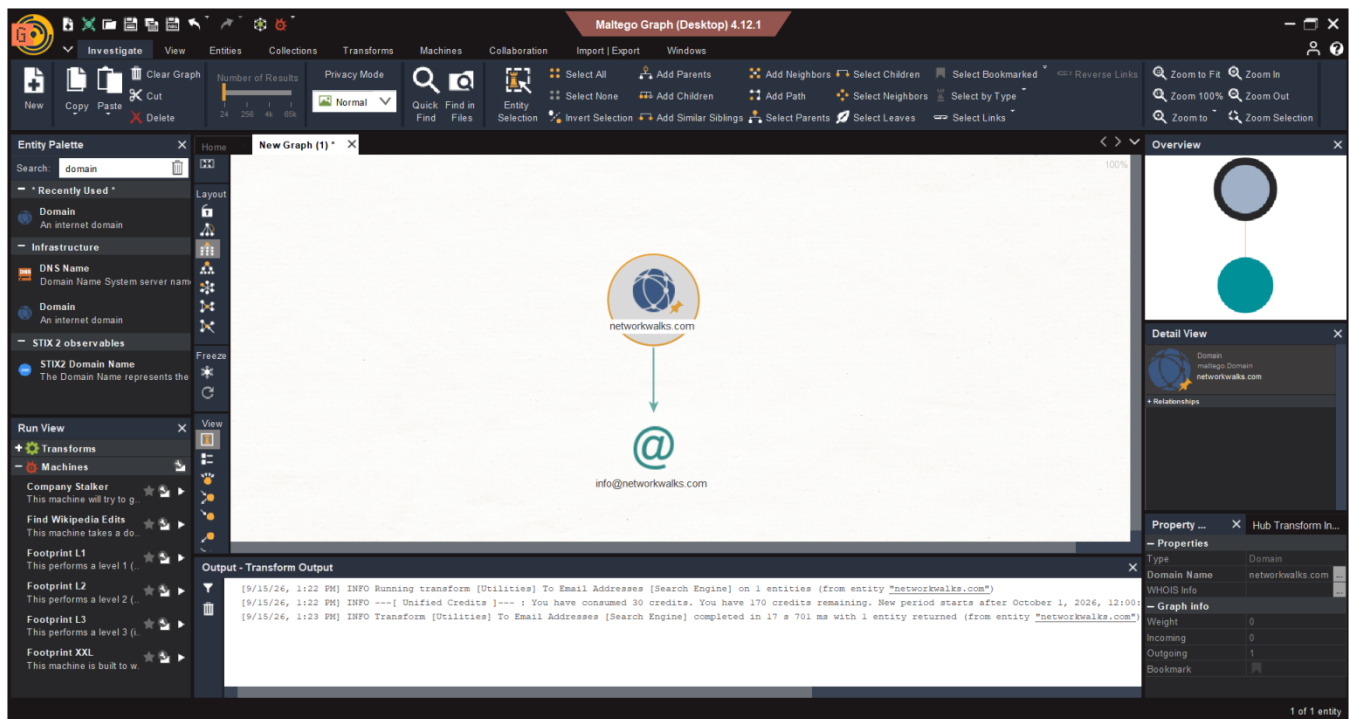


Figure 6 – W2-PM4: theHarvester Baidu results.

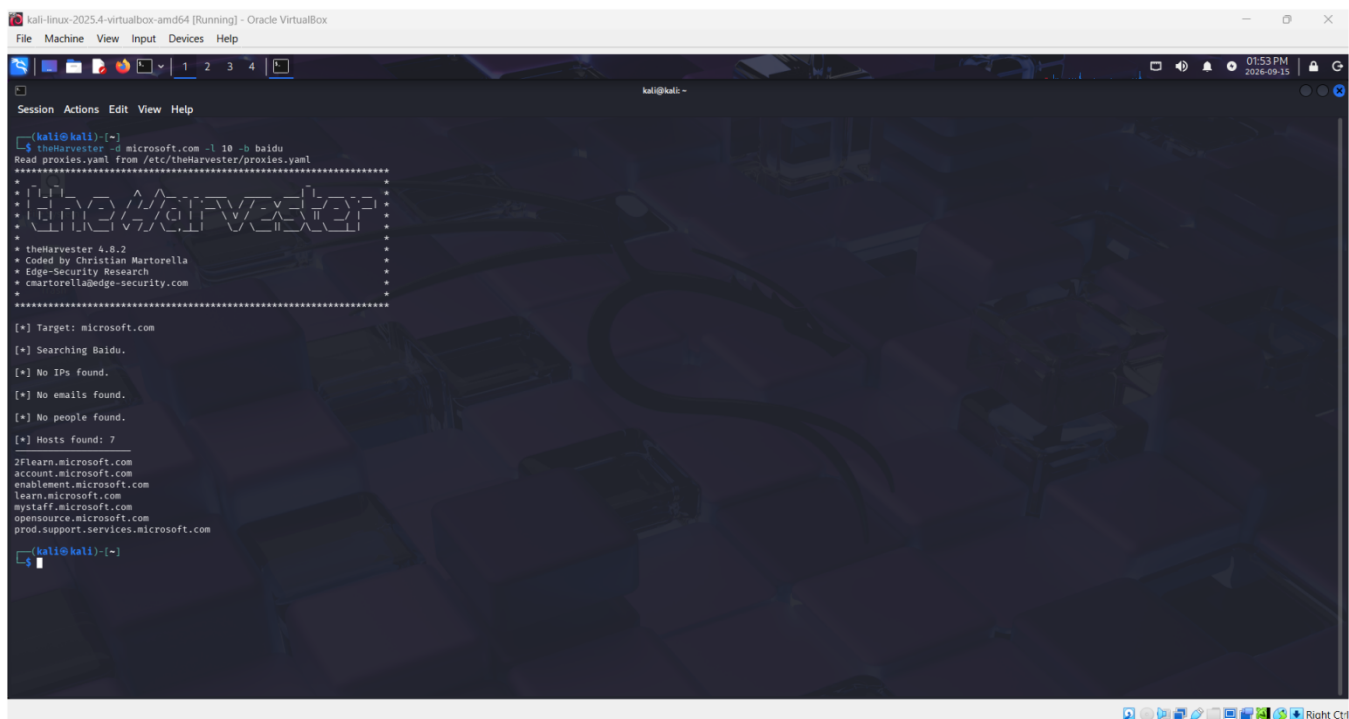


Figure 7 – W2-PM4: theHarvester all-source results and limitations.

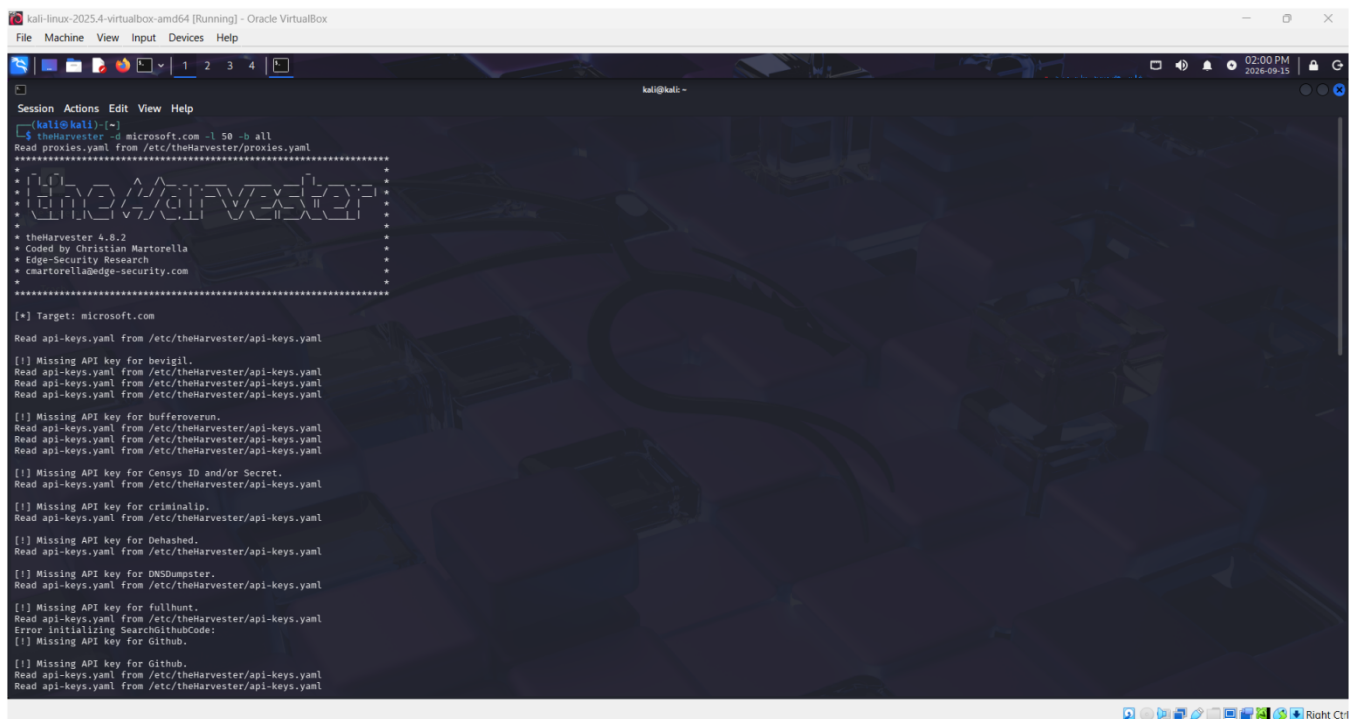


Figure 8 – W2-PM5: Zenmap Ping Scan results showing the four live hosts.

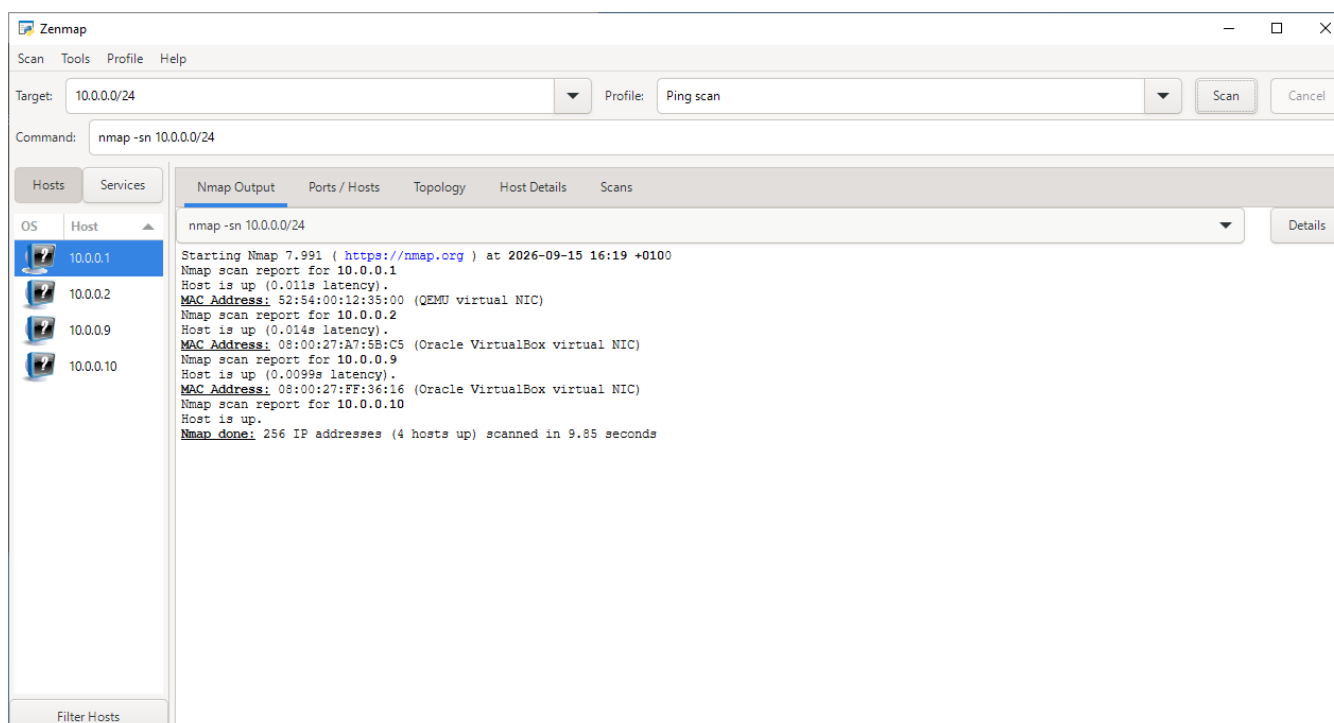
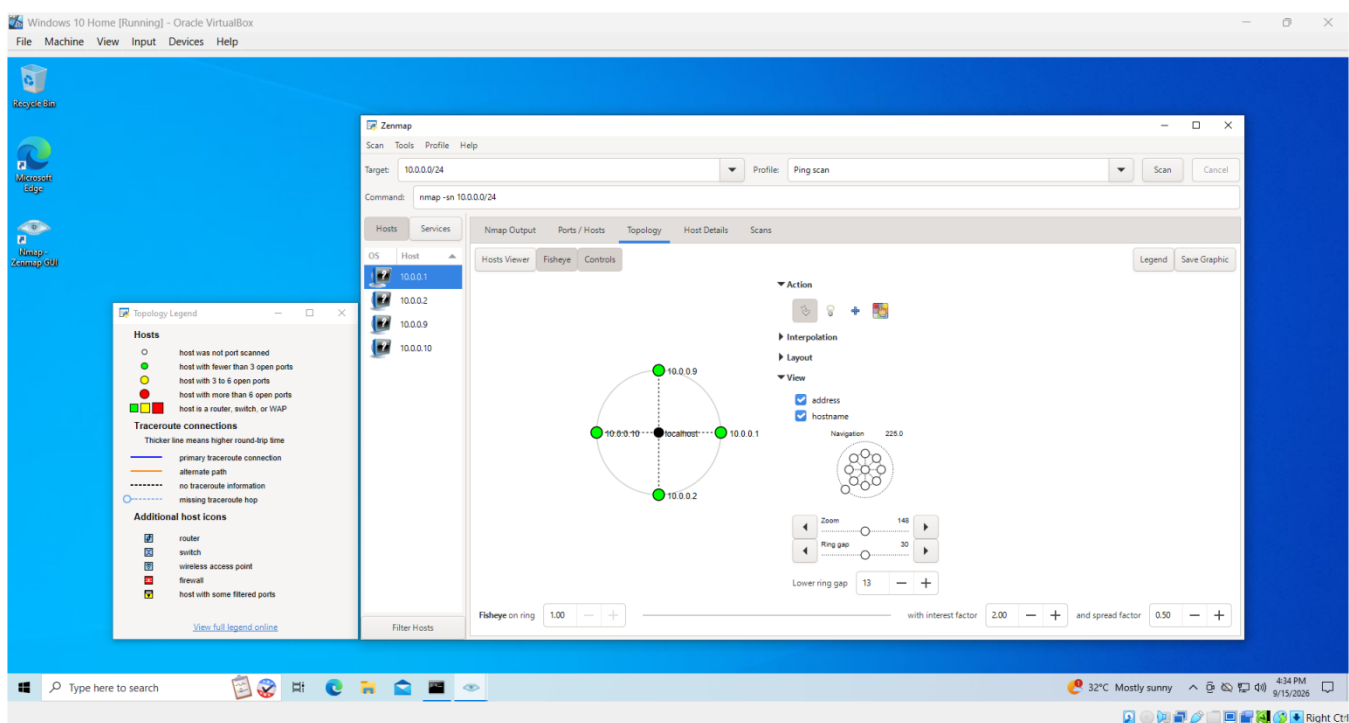


Figure 9 – W2-PM5: Zenmap Topology view and saved PDF evidence.



-End-

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▮ **Project Information**

Program Name: Cybersecurity Internship program at Networkwalks | Week: 02 | Report: W2-PM1-5-FINAL